

Student Name: _____

Class Teacher: _____



HURLSTONE AGRICULTURAL HIGH SCHOOL

2021 Year 12
HSC Task 2

Mathematics Extension 1

Examiners Mr G Rawson, Mrs S Faulds, Mr D Potaczala

General

Instructions

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black or blue pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- An A4 hand written summary sheet may be used.
- In Questions 4-6, show relevant mathematical reasoning and/or calculations.

Total

marks: 33

Section 1 – 3 marks

- Attempt Questions 1-3
- Allow about 4 minutes for this section

Section 2 – 30 marks

- Attempt Questions 4-6
- Allow about 36 minutes for this section

Written by 2021 HAHS Maths Faculty, rewritten by Ariq Abdullah

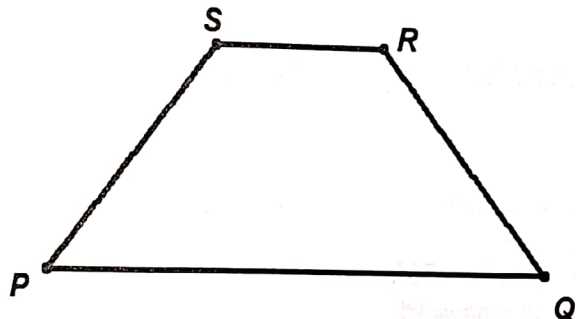
Section 1

Attempt Questions 1-3

Allow about 4 minutes from this section.

Question 1

PQRS is a trapezium where $\overrightarrow{PS} = \vec{p}$, $\overrightarrow{SR} = \vec{s}$, $\overrightarrow{PQ} = 2\overrightarrow{SR}$



Which of the following is equivalent to $\overrightarrow{QS}$?

a. $2\vec{s} + \vec{p}$

c. $\vec{p} - 2\vec{s}$

b. $2\vec{s} - \vec{p}$

d. $-\vec{p} - 2\vec{s}$

Question 2

If A and B are points defined by position vectors $\vec{a} = \vec{i} + \vec{j}$ and $\vec{b} = 5\vec{i} - 2\vec{j}$ respectively, then $|\overrightarrow{AB}|$ is equal to which of the following?

a. $\sqrt{2}$

c. $\sqrt{29}$

b. 5

d. 10

Question 3

What is the number of solutions for the equation $\sin 2x - \sin x = 0$ over the domain $[0, 2\pi]$?

a. 2

c. 4

b. 3

d. 5

Section 2

30 marks

Attempt Questions 4-6

Allow about 36 minutes for this section

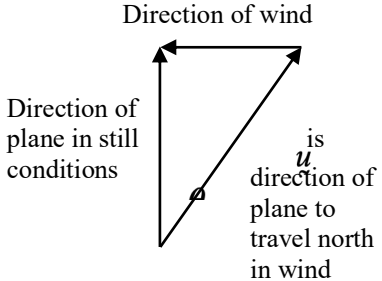
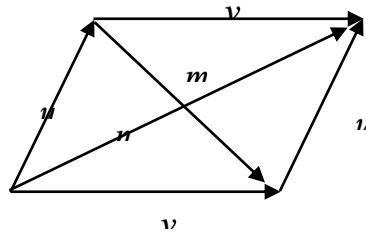
Question 4 (10 Marks)	Marks
(a) Use the principle of Mathematical Induction to show that for all integers $n \geq 1$, $1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n + 1)! - 1$	3
(b)	
i) Show that $(k + 3)^3 = k^3 + 9k^2 + 27k + 27$	1
ii) Using the result in part (i), or otherwise, prove by the principle of Mathematical Induction that the sum of the cubes of three consecutive positive integers is divisible by 3.	3
(c) Prove the following identity: $\frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A} = \tan 2A$	3
Question 5 (10 marks)	Marks
(a) Suppose $ \vec{v} = 3$ and $ \vec{w} = 2$. If $\vec{v}$ is parallel to $\vec{w}$, what values might $\vec{v} \cdot \vec{w}$ take? Give reasons for your answers(s)	2
(b) Given the position vectors $\vec{u} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ Find the vector projection of $\vec{u}$ onto $\vec{v}$.	3
(c) A small plane can fly at 350 km/h in still conditions. The plane needs to fly due north but has to deal with 70km/h winds from due east. Find the direction and speed that the plane needs to fly in order to travel due north (both correct to two decimal places).	2
(d) Prove using vector methods that the sum of the squares of the lengths of the diagonals of a parallelogram is equal to the sum of the squares of the lengths of the sides.	3
Question 6 (10 marks)	Marks
(a) Use the compound angle results to prove the identity $\sin 3\theta - 3\sin\theta - 4\sin^3\theta$	2
(b)	
(i) Express $\cos x + \sqrt{2}\sin x$ in the form $R\cos(x - \alpha)$.	2
Question 6 continued on next page	

- (ii) Hence, solve the equation $\cos x + \sqrt{2}\sin x = \sqrt{3}$, $0 \leq x \leq 2\pi$. 2
- (c) Use the substitution $t = \tan x$ to find all angles x , where $0 \leq x \leq 2\pi$, for which $\sin 2x + \cos 2x = 1$. 2
- (d) Show that the exact value of $\cos\left(\frac{7\pi}{12}\right) = -\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)$. 2

End of Exam

Year 12	Mathematics Extension 1 2021	TASK 2
Question No. 1-4	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
ME12-1 - applies techniques involving proof or calculus to model and solve problems ME12-3 - applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations.		
Part / Outcome	Solutions	Marking Guidelines
(a) ME12-1	Multiple Choice (Q1-3) 1 – C 2 – B 3 – D <u>Show true for $n = 1$</u> LHS = $1 \times 1! = 1$ RHS = $(1+1)! - 1 = 1$ $\therefore$ true for $n = 1$ <u>Assume true for $n = k$</u> ie $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! = (k+1)! - 1$ and <u>Prove true for $n = k + 1$</u> ie $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! + (k+1) \times (k+1)! = (k+2)! - 1$ LHS = $\underbrace{1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k!}_{\text{assumption}} + (k+1) \times (k+1)!$ = $(k+1)! - 1 + (k+1) \times (k+1)!$ = $(k+1)! [1 + (k+1)] - 1$ ie = $(k+1)! (k+2) - 1$ = $(k+2)! - 1$ = RHS $\therefore$ true by mathematical induction for all $n \geq 1$	3 marks – Correct solution 2 mark – Substantially correct 1 mark – Partial progress towards correct solution
(b) (i) ME12-1	$(k+3)^3 = (k+3)(k+3)^2$ = $(k+3)(k^2 + 6k + 9)$ = $k^3 + 6k^2 + 9k + 3k^2 + 18k + 27$ = $k^3 + 9k^2 + 27k + 27$	1 mark – Correct solution

(b) (ii) ME12-1	<i>Question 6 continued...</i> Let the 3 consecutive numbers be $n, n + 1, n + 2$ So, prove $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 3. <u>Show true for $n = 1$</u> $1^3 + (1+1)^3 + (1+2)^3 = 1 + 8 + 27$ $= 36 = 3(12)$ ie, divisible by 3 <u>Assume true for $n = k$</u> ie $k^3 + (k+1)^3 + (k+2)^3 = 3M$, for integer M and $\therefore (k+1)^3 + (k+2)^3 = 3M - k^3$ and <u>Prove true for $n = k + 1$</u> $(k+1)^3 + (k+2)^3 + (k+3)^3 = 3N$, for integer N LHS = $(k+1)^3 + (k+2)^3 + (k+3)^3$ $= 3M - k^3 + (k+3)^3$ <i>from assumption</i> $= 3M - k^3 + k^3 + 9k^2 + 27k + 27$ <i>from (i)</i> $= 3M + 9k^2 + 27k + 27$ $= 3(M + 3k^2 + 9k + 9)$ $= 3N$, where N is an integer $\therefore$ true for $n \geq 1$ by Mathematical Induction	3 marks – Correct solution 2 mark – Substantially correct 1 mark – Partial progress towards correct solution
(c) ME12-3	LHS = $\frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A}$ $= \frac{\sin A(\cos A - \sin A) + \sin A(\cos A + \sin A)}{\cos^2 A - \sin^2 A}$ $= \frac{\sin A \cos A - \sin^2 A + \sin A \cos A + \sin^2 A}{\cos^2 A - \sin^2 A}$ $= \frac{2 \sin A \cos A}{\cos^2 A - \sin^2 A}$ $= \frac{\sin 2A}{\cos 2A}$ $= \tan 2A$ $= \text{RHS}$	3 marks – Correct solution 2 mark – Substantially correct 1 mark – Partial progress towards correct solution

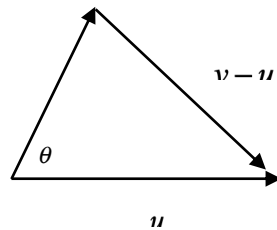
on 1 Task 2 2021		
Question No. 5		
Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
ME12-2 applies concepts and techniques involving vectors and projectiles to solve problems		
Outcome	Solutions	Marking Guidelines
ME12-2	(a) If parallel the angle between vectors is 0° or 180° $\underline{v} \cdot \underline{u} = 3 \times 2 \cos 0 \quad \text{or} \quad \underline{v} \cdot \underline{u} = 3 \times 2 \cos 180$ $= 6 \quad \quad \quad = -6$	2 marks Correct solution 1 mark Substantial progress towards correct solution
ME12-2	(b) $\underline{u} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}; \quad \underline{v} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ $ \underline{v} ^2 = 2^2 + 5^2 \quad \underline{v} \cdot \underline{u} = (-4) \times 2 + 3 \times 5$ $= 5 \quad \quad \quad = 7$ $\frac{\underline{u} \cdot \underline{v}}{ \underline{v} ^2} \cdot \underline{v} = \frac{7}{29} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} \frac{14}{29} \\ \frac{35}{29} \end{pmatrix}$	3 marks Correct solution 2 marks Single error 1 mark Substantial progress towards correct solution
ME12-2	(c) $ \underline{u} = \sqrt{350^2 + 70^2}$ $= 356.93 \text{ km/h}$ $\tan \theta = \frac{70}{350}$ $\theta = 11.31$ $N 11.31^\circ E$ or 011° as a true bearing 	2 marks Correct solution 1 mark Substantial progress towards correct solution Note: marks were still awarded if students interpreted the wind blowing in the easterly direction
ME12-2	(d)  To prove: $2 \underline{v} ^2 + 2 \underline{u} ^2 = \underline{m} ^2 + \underline{n} ^2$ $\underline{m} = \underline{v} - \underline{u} \quad \text{and} \quad \underline{n} = \underline{v} + \underline{u}$	3 marks Correct solution, including correct use of the definition $ \underline{a} ^2 = \underline{a} \cdot \underline{a}$ 2 marks Substantial progress towards correct solution 1 mark Establishes worded question in terms of algebraic vector notation

$$\begin{aligned}
 RHS &= |\underline{m}|^2 + |\underline{n}|^2 \\
 &= |\underline{v} - \underline{u}|^2 + |\underline{v} + \underline{u}|^2 \\
 &= (\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u}) + (\underline{v} + \underline{u}) \cdot (\underline{v} + \underline{u}) \\
 &= |\underline{v}|^2 - 2\underline{v} \cdot \underline{u} + |\underline{u}|^2 + |\underline{v}|^2 + 2\underline{v} \cdot \underline{u} + |\underline{u}|^2 \\
 &= 2|\underline{v}|^2 + 2|\underline{u}|^2 \\
 &= LHS
 \end{aligned}$$

Note: a large portion of students misunderstood how to rewrite $|\underline{v} - \underline{u}|^2 + |\underline{v} + \underline{u}|^2$. Students used binomial expansion to expand the prior expression in all manners of wrong ways to get something similar to $|\underline{v}|^2 - 2\underline{v} \cdot \underline{u} + |\underline{u}|^2 + |\underline{v}|^2 + 2\underline{v} \cdot \underline{u} + |\underline{u}|^2$.

The pike symbols (| |) do not behave like brackets and thus binomial expansion does not apply. Nor does it really apply to $(\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u})$ (just try $(\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u})$).

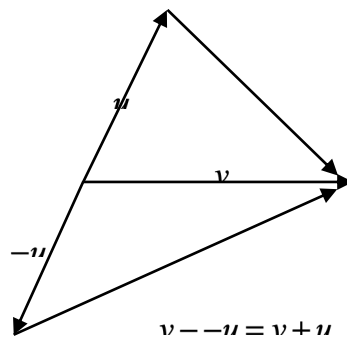
What should have been used is the definition $|\underline{a}|^2 = \underline{a} \cdot \underline{a}$. From there you get $|\underline{v} - \underline{u}|^2 = (\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u})$. To understand how the dot product can be “expanded” you can use the cosine rule.



$$\begin{aligned}
 \text{It is clear that } |\underline{v} - \underline{u}|^2 &= |\underline{v}|^2 + |\underline{u}|^2 - 2|\underline{v}||\underline{u}|\cos\theta \\
 &= |\underline{v}|^2 + |\underline{u}|^2 - 2\underline{v} \cdot \underline{u} \quad (\text{dot product})
 \end{aligned}$$

$$\text{This means } (\underline{v} - \underline{u}) \cdot (\underline{v} - \underline{u}) = |\underline{v}|^2 + |\underline{u}|^2 - 2\underline{v} \cdot \underline{u}$$

$$\begin{aligned}
 \text{Similarly, } |\underline{v} + \underline{u}|^2 &= |\underline{v}|^2 + |-\underline{u}|^2 - 2|\underline{v}||-\underline{u}|\cos(180 - \theta) \\
 &= |\underline{v}|^2 + |\underline{u}|^2 + 2\underline{v} \cdot \underline{u} \\
 &= (\underline{v} + \underline{u}) \cdot (\underline{v} + \underline{u})
 \end{aligned}$$



Year 12 Mathematics Extension 1 Task 2 2021		
Question No. 6 Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
ME12.3 – applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations/		
Outcome	Solutions	Marking Guidelines
ME12.3	(a) $\begin{aligned} \text{LHS} &= \sin 3\theta \\ &= \sin(2\theta + \theta) \\ &= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\ &= (2 \sin \theta \cos \theta) \cos \theta + (\cos^2 \theta - \sin^2 \theta) \sin \theta \\ &= 2 \sin \theta \cos^2 \theta + \sin \theta \cos^2 \theta - \sin^3 \theta \\ &= 3 \sin \theta \cos^2 \theta - \sin^3 \theta \\ &= 3 \sin \theta (1 - \sin^2 \theta) - \sin^3 \theta \\ &= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \\ &= \text{RHS} \end{aligned}$	2 marks Correct solution. 1 mark Substantial progress towards correct solution showing knowledge of some of the appropriate trig, identities.
ME12.3	(b) (i) $R = \sqrt{1^2 + (\sqrt{2})^2} = \sqrt{3}$ $\cos x + \sqrt{2} \sin x = \sqrt{3} \left(\frac{1}{\sqrt{3}} \cos x + \frac{\sqrt{2}}{\sqrt{3}} \sin x \right)$ $= \sqrt{3} (\cos(x - \alpha))$ where $\alpha = \tan^{-1} \sqrt{2}$ $= 0.955$ $\therefore \cos x + \sqrt{2} \sin x = \sqrt{3} (\cos(x - 0.955))$	2 marks Correct solution giving values for both R and α. 1 mark Substantial progress towards correct solution. Note: Answer for α is more correctly stated in radians. Marks were also awarded for equivalent answer stated in degrees/
ME12.3	(ii) $\begin{aligned} \cos x + \sqrt{2} \sin x &= \sqrt{3} & 0 \leq x \leq 2\pi \\ \sqrt{3} (\cos(x - 0.955)) &= \sqrt{3} \\ \cos(x - 0.955) &= 1 & -0.995 \leq x - 0.995 \leq 2\pi - 0.995 \\ (x - 0.955) &= \cos^{-1} 1 \\ &= 0 \\ \therefore x &= 0.995 & 0 \leq x \leq 2\pi \end{aligned}$	2 marks Correct solution giving a single value for x in radians. 1 mark Substantial progress towards correct solution.

ME12.3

(c)

$$\sin 2x + \cos 2x = 1$$

$$\text{Let } t = \tan \frac{2x}{2} = \tan x$$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1 \quad 0^\circ \leq x \leq 360^\circ$$

$$2t + 1 - t^2 = 1 + t^2$$

$$= 2t^2 - 2t$$

$$= 2t(t-1)$$

$$t = 0, 1$$

$$\text{ie. } \tan x = 0, 1$$

$$x = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$$

2 marks

Correct solution giving all correct values for x in radians.

1 mark

Substantial progress towards correct solution.

ME12.3

(d)

$$\cos\left(\frac{7\pi}{12}\right) = \cos\left(\frac{3\pi}{12} + \frac{4\pi}{12}\right)$$

$$= \cos\left(\frac{\pi}{4} + \frac{\pi}{3}\right)$$

$$= \cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right) - \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{2} - \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$$

$$= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{1-\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{\sqrt{2}-\sqrt{6}}{4}$$

$$= -\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \quad \text{as required}$$

2 marks

Correct solution showing all required steps.

1 marks

Substantial progress towards correct solution, showing most of the required steps including knowledge of relevant trig. identities and exact ratios.